Supplement to: A Reproducible Multi-pollutant Railway Emissions Inventory for Pakistan with Fuel-Balance Closure and Route-Constrained Spatial Allocation

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Abstract

Generated registers and audits behind the main text: the uncertainty inputs, the full structural-scenario table, the CRT cell audit, the Tier 2 rescaling, the network audit by corridor, the sulfur ladder, and the quinquennial fuel history. Every table here is a rendering of a versioned CSV in `outputs/`; the CSVs are authoritative.

Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc69`
Commit	`f3f7faec9d6a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`c5ca5795eb24666c` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. The sulfur ladder

Case	Diesel S (ppm)	SO_2 (kt)
euro5_import_spec	10	0.0078
reference_pre_euro5_spec	500	0.1176
historical_0.5pct	5000	1.1260
historical_1.0pct	10000	2.2463

Table 1: SO_2 under each declared sulfur case (stoichiometric, so the case is the answer); the furnace-oil floor of 0.0056 kt is common to all rows. From `railways_air_sulfur_cases_2024.csv`.

2. The Tier 2 rescaling

Service	kt diesel	share (%)
line-haul passenger	86.393	77.11
line-haul freight	20.042	17.89
shunting	5.602	5.00

Table 2: The service decomposition, rescaled exactly to the 112.04 kt control (asserted in-function and by test). Railcars are not occurring. The shunting share is a declared assumption; see `rail-service-split` in the gap register. From `railways_tier2_split_2024.csv`.

3. The network audit by corridor

Corridor	km
ML1_south (Karachi–Rohri)	476.4
ML1_central (Rohri–Lahore)	787.2
ML1_north (Lahore–Peshawar)	459.3
ML2 (Kotri–Kundian)	1052.7
ML3 (Rohri–Quetta)	328.4
BR_faisalabad	302.4
BR_sialkot	41.6
BR_sargodha	217.4
branch_other	3908.9

Table 3: Active network by corridor; 7574 km in total against the Year Book's 7791 route-kilometres (ratio 0.972), with 1246 km of disconnected fragments carrying no line-haul weight. From `railways_network_audit_2024.csv`.

4. The quinquennial history

Period	Coal (t)	FO (t)	HSD (kL)	Electric (MWh)	~Mt CO_2e
1950–55	321412	449555	7785		2.16
1955–60	173092	494133	35829		2.12
1960–65	103615	521294	80891		2.22
1965–70	48720	542078	118500		2.30
1970–75	27118	374105	128335		1.72
1975–80	6273	310151	141486		1.50
1980–85	2547	257056	148603		1.33
1985–90	1965	124229	167212	37402	0.93
1990–95	1107	52464	151631	27918	0.63
1995–2000	541	14490	137631	10640	0.46
2000–05	89	3423	134509	5969	0.41
2005–10	108	1628	146134	6133	0.44
2010–15	40	410	103099	0	0.31
2015–20	0	454	148008	0	0.44

Table 4: Fuel per year (five-year averages) with an indicative CO_2e conversion; the structural breaks—the 1971 boundary change, the end of steam by 1980–85, the electric section's reported life 1985–90 to 2005–10—are annotated row by row in the activity file. Annual figures from FY2014 live in the inventory series. From `railways_fuel_history.csv`.

5. The CRT cell audit

The extraction re-derives every value from the archived workbook (`PAK_CRT_2024.zip`, SHA-256 pinned by test)

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on each run, rows located by label so a workbook revision breaks loudly. The category row books 56 609.03 TJ (NCV basis), 4194.75 kt CO_2, 0.2208 kt CH_4 and 0.2207 kt N_2O; the liquid-fuels row's implied-factor cells read 74.10 t CO_2/TJ, 3.90 kg CH_4/TJ and 3.8995 kg N_2O/TJ. The full cell-by-cell listing, with workbook coordinates, is `railways_unfccc_crt_cell_audit_2024.csv`.

6. The uncertainty inputs register

Every quantity the uncertainty statement rests on is a row of `railways_uncertainty_inputs_2024.csv`: parametric rows (fuel quantities, density, calorific values, gas and air factors, the BC fraction) each with source, distribution and correlation group; structural rows (the factor-table choice, the 1A3c boundary, consumed-equals-burnt timing, the sulfur case) explicitly *not sampled*; and one row marked RETIRED—the summary-vs-detail diesel conflict, settled by the 2024–25 edition's restatement—kept so its absence from the scenario table reads as resolution rather than omission.